Xuanyi Chen

(+86) 185-3799-4689 • EDU EMAIL • GMAIL • PERSONAL WEBSITE

RESEARCH PROFILE

Aspiring plant scientist interested in the mechanisms by which plants sense and respond to environmental stresses, combining molecular genetics, cellular biology, and integrative multi-omics analyses.

EDUCATION

PhD in Genetics (Plant Molecular Genetics) **Sep 2021 – Dec 2025**

Northwest A&F University, Yangling, China

Dissertation: Molecular mechanism of BR signaling pathway of regulating uptake and utilization of sulfur in *Arabidopsis thaliana*

Advisor: Professor Cun Wang

BSc in Biotechnology (Honors Diploma) **Sep 2017 – Jun 2021**

Northwest A&F University, Yangling, China

RESEARCH EXPERIENCE

Graduate Research Fellow **Jun 2021 – Dec 2025**

Advisor: Professor Cun Wang

• Investigating molecular mechanisms underlying plant sulfur-deficiency responses, focusing on brassinosteroid and calcium signaling pathways.

• Integrated molecular genetics, biochemistry, cell biology, and multi-omics approaches.

Undergraduate Research Assistant – Epigenome Editing **Jun 2020 – Jun 2021**

Advisor: Professor Israel Ausin

• Designed zinc finger–tagged SWR1 chromatin remodeling complex subunits to profile protein–DNA interactions.

Undergraduate Research Assistant – Virome Analysis **Sep 2018 – Jun 2020**

Advisor: Professor Weimin Chen

• Characterized the rhizosphere virome of *Robinia pseudoacacia* L. and isolated bacteriophage strains.

Undergraduate Research Assistant – Bioinformatics Training **Sep 2017 – Jun 2018**

Advisor: Professor Ruolin Yang

• Performed TCGA data mining and received foundational training in bioinformatics.

GRANT

Participant, National Natural Science Foundation of China (NSFC) **2021-2025**

Principal Investigator: Professor Cun Wang
Total funding: 2,000,000 CNY (~290,000 USD)

Project Leader, Undergraduate Training Program for Innovation **2019–2021**
Supervised by: Professor Weimin Chen
Total funding: 3,500 CNY (~500 USD)

PUBLICATIONS

Chen X., Yu Z., Guo W., Zhou Y., Wang C.#, Wang T.# (2025)
Brassinosteroid signaling promotes sulfate uptake under sulfur deficiency in Arabidopsis.
New Phytologist 248: 250–264.

Fang Y.*, Ju C.*, Javed L.*, Cao C., Deng Y., Gao Y., **Chen X.**, Sun L., Zhao Y., Wang C.# (2025)
Plasma membrane–associated calcium signaling modulates zinc homeostasis in *Arabidopsis*.
Science Bulletin 70: 1478-1490.

Jia M., **Chen X.**, Shi X., Fang Y., Gu Y.# (2023)
Nuclear transport receptor KA120 regulates molecular condensation of MAC3 to coordinate plant immune activation.
Cell Host & Microbe 31: 1685–1699.e7.

Wang T., **Chen X.**, Ju C.#, Wang C.# (2023)
Calcium signaling in plant mineral nutrition: From uptake to transport.
Plant Communications 4: 100678. (Highly cited review)

Chen X., Li S.#, Chen W., Meng Q. (2021)
Characters of light-sheet fluorescence microscope and its application.
Shengwujishu Jinzhan 11: 126–147. (Review in Chinese)

AWARDS & HONORS

National Scholarship for Doctoral Students	2025
First-Class Scholarship	2023
Second-Class Scholarship	2021, 2022, 2024
Advanced Undergraduate of Technological Innovation	2021

PROFESSIONAL SERVICE

Assistant Student Editor **Jul 2023 – Dec 2025**
Stress Biology
• Managed the journal's X (Twitter) and Bluesky social media platforms
• Assisted with the journal's participation in three academic conferences

• Processed and responded to email inquiries from Chinese authors

TECHNICAL SKILLS

Cell Biology: Confocal microscopy, BiFC assays.

Molecular Biology: qPCR, mutagenesis, ^{32}P labeling of protein phosphorylation analysis, Western blot, ChIP, EMSA, IP, reporter assays, CRISPR-based gene editing.

Plant Biology: Arabidopsis and wheat transformation, Arabidopsis hybridization, element content analysis.

Bioinformatics: R (proficient), Python (basic), RNA-seq, ChIP-seq, proteomics, alternative splicing, AlphaFold analysis.